

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 6: Solving Problems Based on Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems on speed, distance and time

1. A couple drive 350 km to a hill station. Heavy rain reduces their average speed by 20 km/h, and the journey takes 2 hours longer. Find (a) the original speed of the car (b) the time taken at the reduced speed. **ICSE 2025, 4 marks**

SOLUTION

(a) Let the original speed be x km/h. $\frac{350}{x-20} - \frac{350}{x} = 2$

$350x - 350(x-20) = 2x(x-20)$, so $7000 = 2x^2 - 40x$ and $x^2 - 20x - 3500 = 0$

$(x-70)(x+50) = 0$, so $x = 70$ (speed cannot be negative).

(b) Reduced speed = 50 km/h, so the time is $\frac{350}{50} = 7$ hours.

Answer: (a) 70 km/h (b) 7 hours

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